

Problem 25.13

A bank makes payments continuously at a rate of \$400 a year. The payments are made between 5 and 7 years. Find the current value of these payments at time 2 years using an annual rate of discount of 4%.

Solution

The annual effective rate of discount is

$$d = 0.04.$$

Since

$$d = \frac{i}{1+i},$$

we have

$$0.04 = \frac{i}{1+i}.$$

Thus,

$$0.04(1+i) = i,$$

so

$$0.04 = 0.96i.$$

Therefore,

$$i = \frac{0.04}{0.96} = \frac{1}{24}.$$

The discount factor is

$$v = 1 - d = 0.96.$$

The payments are made continuously from time 5 to time 7, a period of 2 years. The value at time 5 of these payments is

$$400\bar{a}_{\overline{2}|}.$$

To find the value at time 2, discount this value for 3 years:

$$PV_2 = 400\bar{a}_{\overline{2}|}v^3.$$

For a continuous annuity,

$$\bar{a}_{\overline{2}|} = \frac{1-v^2}{\delta},$$

where

$$\delta = \ln(1+i).$$

Since $v = 0.96$,

$$\delta = -\ln(v) = -\ln(0.96) = 0.0408219945.$$

Thus,

$$PV_2 = 400 \left(\frac{1 - (0.96)^2}{0.0408219945} \right) (0.96)^3.$$

Therefore,

$$PV_2 = 679.6659812.$$

So the current value at time 2 is

$$\boxed{679.67}.$$